Week 3: proof by contradiction

Prove that for $x, y \in \mathbb{R}$,

$$x + y > 20 \Rightarrow \boxed{x > 10 \text{ or } y > 10.}$$

P

Proof: Assume $x + y > 20$. Assump.

(Q is either true or false)

(either "Q" or $\neg Q$ is true)

Proceed by contradiction. Assume $\neg P$, i.e., $x \leq 10$ <u>and</u> $y \leq 10$. [Adding gives $x+y \leq 20$.] Q This contradicts our assumption. Thus $\neg P$ must have been false, so P is true, i.e.,

$x > 10$ or $y > 10$. ∎

Proof that $\neg P \Rightarrow Q$

Template for proof by contradiction

- We want to prove P.
- There are "2 cases": P is true or P is false.
- If we can rule out "P is false", then P must be true.
- To begin, Assume P is false, i.e. $\neg P$.
- "Argue"; i.e. exhibit some (correct) chain of reasoning. end up w/ a statement Q.

(I.E write out a proof of "$\neg P \Rightarrow Q$")

- Observe (or give a proof that) Q is false.
- Conclude that $\neg P$ is false.

$$(\neg P \Rightarrow Q) \wedge \neg Q \Rightarrow P$$

Intuition:

- Chess "If I move here, in 4 moves they have check. So I shouldn't make that move"

Prove that $x^2 - y^2 = 1$ has no positive integer solutions.

$(x,y) = (1,0)$

not positive.

Proof: Proceed by contradiction. Assume that x, y is a positive ? integer solution to $x^2 - y^2 = 1$. Then $(x-y)(x+y) = 1$. There are 2 possibilities:

$x - y = 1$ and $x + y = 1$, or $x - y = -1$ and $x + y = -1$.

In the 1st case adding gives $2x=2$
Then $x=1$, but $x-y=1$, $\boxed{y=0}$. Q_1 this is a contradiction since y is positive,
In the 2nd case, adding gives $2x=-2$,
So $\boxed{x=-1}$. Q_2 This is a contradiction since x is positive.

In both cases, we get a contradiction. Thus our assumption was wrong, and we conclude that there are no positive integer solutions to $x^2 - y^2 = 1$. $\blacksquare$

N.B. Really was a proof that

$$\neg P \Rightarrow (Q_1 \text{ or } Q_2)$$

need both to give a contradiction

IE if there are cases in conclusion, need each case to be false.

Prove that $x^2 = 4y + 3$ has no integer solutions. P

Proof: Proceed by contradiction. Assume that there are $x, y \in \mathbb{Z}$ s.t.

$$x^2 = 4y + 3.$$

There are 2 cases: x is even or x is odd.
If x is even then the LHS is even
and the RHS is odd. This is a
contradiction.
If x is odd, then $x = 2k+1$ for some
$k \in \mathbb{Z}$. Plugging in gives

$$(2k+1)^2 = 4k^2+4k+1 = 4y+3.$$

This a contradiction, since the LHS has a remainder of 1 and the RHS has a remainder of 3. $\blacksquare$

Recall: An integer n is prime if its only divisors are ± 1 and $\pm n$.

$2, 3, 5, 7, 11, 13, 17, 19, 23,$ primes

$4, 6, 8, 12$ not prime

$65537 = 2^{2^4} + 1$

65538 not prime

Euclid's theorem: There exist infinitely many primes.

Proof: Proceed by contradiction. Assume there are only finitely many primes.
Let's name them $p_1, p_2, \ldots, p_r$.

$(p_1 = 2, p_2 = 3, p_3 = 5, \ldots \; p_r = ?)$

Label them so that $p_1 = 2$, $p_{i+1} > p_i$

let $N = p_1 p_2 p_3 \cdots p_r + 1$.

(what does the factorization look like?)

Since $N > p_r$, N isn't prime (b/c N is bigger than the biggest prime). By FTA (Fundamental thm of arithmetic), N

factor into primes. let q be one of those primes. Since $p_1, \ldots, p_r$ are all of the primes, $\exists i$ s.t. $q = p_i$.

then $q \mid N$ and $q \mid p_1 \cdots p_i \cdots p_r$, thus

by the proof of 3 rules,

$q \mid N - (p_1 \cdots p_r)$, i.e. $q \mid 1$. This is a contradiction since $q > 1$.

Thus there are infinitely many primes. $\blacksquare$

__Joke:__ There are no uninteresting ~~numbers~~. positive integers.

__Proof:__ Proceed by contradiction. Assume that some positive integers are uninteresting. Then there must be a smallest uninteresting positive integer.

But that's pretty intresting!

(this is a proof technique: think about the "smallest" counterexample) if $a|b$ and $a|b+c$ then $a|c$

$a|-b$

$a \mid (b+c)+(-b) = c$

FTA (Fundamental THM of Arithmetic)

let $N \in \mathbb{Z}_{\geq 1}$. Then

(i) $\exists\ p_1, \ldots, p_r$ primes s.t.

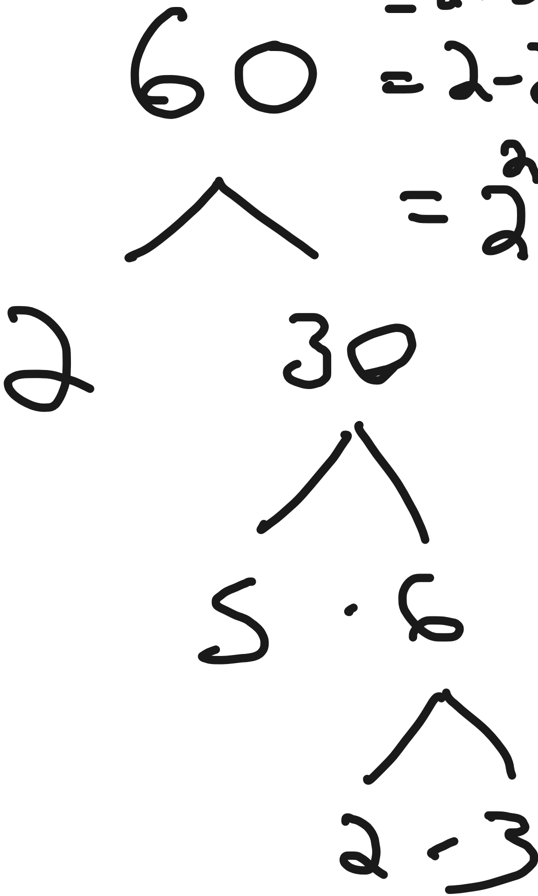
$$N = p_1 \cdot \cdots p_r$$

(ii) If $N = p_1 \cdots p_r = q_1 \cdots q_s$ s.t.

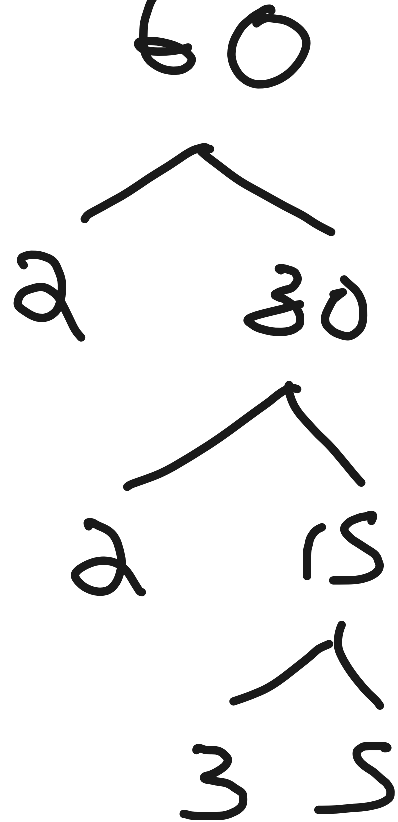
$p_i \geq p_{i-1}$ and $q_j \geq q_{j-1}$, then $r = s$

and $\forall i,\ p_i = q_i$

$$= 2 \cdot 5 \cdot 2 \cdot 3$$
$$= 2 \cdot 2 \cdot 3 \cdot 5$$
$$= 2^2 \cdot 3 \cdot 5$$



$$2 \cdot 2 \cdot 3 \cdot 5$$



Prove (i) Proceed by contradiction. Assume that some $N \in \mathbb{Z}_{>1}$ do not factor into primes. Let N be the smallest such integer. Then N is not prime, otherwise it is already factored

Thus N is composite. Write $N = ab$ where $1 < a, b < N$.
Since $a, b < N$ and N is the smallest integer that doesn't factor, a and b factor. Write $a = p_1 \cdots p_r$, $b = q_1 \cdots q_s$.
Then $N = ab = (p_1 \cdots p_r)(q_1 \cdots q_s)$.

This is a contradiction, since we just factored N. ∎